PES UNIVERSITY
Department of Computer Science and Engineering
UE24CS352A: MACHINE LEARNING
Lab 5: Linear vs. Logistic Regression and Threshold Tuning

Lab Overview & Objectives

The goal of this assignment is to provide hands-on experience with binary classification by transitioning from linear models to probabilistic models. The provided Jupyter Notebook will be used to explore decision boundaries and classification thresholds.

Learning Goals

- Understand why linear regression is mathematically unsuitable for classification tasks.
- Implement a scikit-learn Pipeline integrating feature scaling, feature selection, and logistic regression.
- Extract raw prediction probabilities using the predict_proba() method.
- Analyze the impact of custom classification thresholds on the Confusion Matrix and evaluate the trade-off between Precision and Recall.

Datasets

- **1D Toy Dataset:** A simple, synthetically generated dataset ("Hours Studied" vs. "Pass/Fail") designed for 2D visualization of decision boundaries.
- **Real-World Dataset:** The Wine Quality dataset, predicting whether a red wine is of 'good quality' based on its chemical properties.

Key Concepts

- **Linear Limitations:** Linear regression outputs continuous values extending to infinity. It cannot strictly map predictions to valid probabilities between 0 and 1, making it highly sensitive to extreme outliers in classification tasks.
- **The Logistic Solution:** Logistic regression utilizes the sigmoid function, $S(z) = \frac{1}{1+e^{-z}}$, to compress continuous outputs into a bounded probability range of [0, 1].
- **Threshold Tuning:** The default classification threshold is 0.5. Modifying this threshold directly shifts the balance between false positives and false negatives, directly impacting Precision and Recall.

Instructions and Tasks

The main requirement is to complete the TODO sections within the provided boilerplate notebook.

Part 1: The Limitations of Linear Regression (Steps 2-4)

The synthetic 1D dataset must be utilized to fit a standard linear regression model.

- Generate the toy dataset using the provided seed and arrays.
- Initialize and fit a LinearRegression model.
- Plot the original data points alongside the linear regression line.
- Identify and document predictions that fall outside the [0, 1] probability range. Observe how the single extreme outlier drastically skews the decision boundary.

Part 2: The Logistic Solution (Steps 5-6)

A logistic regression model must be applied to the exact same 1D Toy Dataset.

- Initialize and fit a LogisticRegression model.
- Extract probability predictions for a dense range of values to plot a smooth sigmoid curve over the scatter plot.
- Compare the S-curve to the linear line, specifically noting how the logistic function resists the pull of the extreme outlier.

Part 3: Complete Pipeline and Threshold Tuning (Steps 7-9)

Transition to the Wine Quality dataset to build a robust classification workflow.

- **Pipeline Setup:** Construct a scikit-learn Pipeline consisting of StandardScaler, SelectKBest (k=8), and a LogisticRegression classifier. Fit this pipeline on the training data.
- **Probability Extraction:** Instead of using the default .predict() method, utilize .predict_proba() on the test set to extract the raw probabilities for the positive class.

- **Threshold Application:** Manually apply three distinct classification thresholds (0.3, 0.5, and 0.7) to the raw probabilities to generate three separate sets of binary predictions.
- **Evaluation:** For each threshold, generate a Confusion Matrix using ConfusionMatrixDisplay. Calculate and log Accuracy, Precision, Recall, and F1-Score.

Expected Deliverables

1. Completed Jupyter Notebook (.ipynb)

- The notebook must be fully executed, with all TODO sections completed and all output cells (printouts, plots) visible.
- Ensure the code is clean, well-commented, and runs without errors from top to bottom.

2. Comprehensive Lab Report (.pdf): A formal report detailing the findings must be submitted. The structure must include:

- **Title Page:** Project Title, Student Name, Student ID, Course Name, Submission Date.
- **Introduction:** A brief overview of the project's purpose.
- **Methodology:** An explanation of the ML pipeline and threshold tuning concepts.
- **Results and Analysis:**
    1. **Visual Comparison:** Include the plots from Part 1 and Part 2. Detail the visual differences between linear and logistic models when handling the outlier.
    2. **Performance Table:** Display the key performance metrics (Accuracy, Precision, Recall, F1-Score) across the three custom thresholds.
    3. **Threshold Trade-offs:** Include the Confusion Matrix plots. Explain the inverse relationship between Precision and Recall as the threshold increases, and propose which threshold is best suited for the Wine Quality context.
    4. **Conclusion:** Summarize key takeaways regarding model selection and probability thresholds.
- **Answers to the following questions**
    1. **Precision vs. Recall:** Observe the evaluation metrics generated across the 0.3, 0.5, and 0.7 thresholds. Explain the mathematical or logical reasoning behind the inverse relationship between Precision and Recall as the decision threshold shifts.
    2. **Contextual Threshold Selection:** In the context of predicting Wine Quality, define what a False Positive and a False Negative represent. Detail a specific business scenario where a lower threshold (e.g., 0.3) would be strategically preferred over a higher threshold (e.g., 0.7).

3. **The Problem with MSE:** Mean Squared Error (MSE) is the standard cost function for linear regression. Explain why applying MSE to a classification model utilizing the sigmoid function is mathematically problematic. Focus the explanation on the resulting shape of the cost surface and the concept of local minima.
4. **Binary Cross-Entropy (Log-Loss):** Logistic regression relies on the Log-Loss cost function. Describe how this function behaves when evaluating predictions, specifically how it mathematically penalizes confidently incorrect predictions compared to marginally incorrect ones.
5. **Gradient Descent Mechanics:** Gradient descent is utilized to find the optimal parameters by minimizing the cost function. Explain why gradient descent operates reliably and efficiently on the Log-Loss cost function in logistic regression. Relate this explanation to the concept of convexity.

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